

What is Excel?

Microsoft Excel is a spreadsheet application used to store, calculate, analysis, & manage data.

Uses :- Data entry, Data calculation, Reports, Charts, Salary Calculation, Attendance, Inventory, Division

Calculation $10 + 10 = 20$

Reports

Charts

Salary Calculation

Attendance

Inventory

Division

Workbook :-

A workbook is an Excel file.

EX: Employee.xlsx

Worksheet :-

A worksheet is a single sheet inside a workbook.

* One workbook can have many worksheets.

Row :- ① Horizontal line

② Number : (1, 2, 3, 4) ...

Column :-

① vertical line

② Lettered : A, B, C, ...

Cell :-

The intersection of a row & a column.

Ex: ① A1, ② B5 ③ C10

Cell Reference :- The address of a cell

Ex: A1, B2, C3, etc.

Formula :- A formula always starts with =

Ex: =10+20

= A1 + B1

Basic Operators :-

① + Addition

② - Subtraction

③ * Multiplication

④ / Division

⑤ ^ Power

⑥ % Percentage

Common Function :-

① Sum :- = Sum (A1:A5)

Adds numbers

② Average :- = Average (A1:A5)

Find the average

③ Max :- = Max (A1:A5)

Largest value

④ Min :- = Min (A1:A5)

Smallest value

⑤ Count :- = Count (A1:A5)

Count numbers

IF Function :-

Syntax :-

= IF (condition, true, false)

EX: = IF (A1 >= 50, "Pass", "Fail")

If marks are 50 or more → Pass, otherwise Fail

Comparison Operators :-

① = Equal

② > Greater than

③ < Less than

④ >= Greater than or equal to

⑤ <= Less than or equal to

⑥ <> Not equal

Autofill :-

Drag the fill handle to copy formulas or continue a series.

Shortcut Key :-

① Ctrl + C → Copy

② Ctrl + V → Paste

③ Ctrl + X → Cut

④ Ctrl + Z → Undo

⑤ Ctrl + Y → Redo

⑥ Ctrl + S → Save

⑦ Ctrl + A → Select all

⑧ Ctrl + B → Bold

⑨ Ctrl + P → Print, F2 → edit cell

Text → Left side
Number → Right
boolean → in Between

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Date _____
Page _____

① Worksheet

worksheet in a single sheet in Excel.

↓ sheet :- Column - A, B, C, D

rows - 1, 2, 3, 4

② Cell :-

A cell is the intersection of a row & a column.

EX: D3 = Column D + Row 3

A1 = Column A + Row 1

③ Formula :-

A Formula is used to Perform calculation in Excel.

Every Formula starts with **=**

EX: = D3 + D2

Sum :- **= sum (A1 + A5)**

④ Text :-

Text means words, names or Places entered in a cell.

EX: D3 = Pune

Text is aligned Left by default

No are aligned Right by default

Logical (boolean) values

① true ② false

⑤ Range :- A range is a group of cell.

Shortcut formula: $C2:C10$

1
2
3
4
5
6
7
8
9
10

Shortcut :- Ctrl + Shift + ↓ ⇒ select all data download.

⑥ Sum function :-

The sum function is used to add No.

$= \text{sum}(A1:A10)$

A	
1	
2	1+2+3
3	+4+5
4	+6+7
5	+8+9+10
6	
7	
8	
9	
10	

⑦ Percentage :-

$$\text{Percentage} = \frac{\text{obtained mark}}{\text{Total mark}} \times 100$$

EX: obtained marks = 450

Total marks = 500

calculation :-

$$\text{Per} = \frac{450}{500} \times 100$$

$$= 90$$

$\text{Per} = 90\%$

Cell Reference :-

A cell reference is the address of a cell in Excel.

- * It tells Excel which cell to use in a formula.

Ex: A1 → Column A, Row 1

Type of Reference

① Relative Cell Reference

(Copy A1) →

- ⇒ It changes automatically when you copy the formula.

⇒ Ex: A1

Ex: A1 B1 C

10 20 = A1 + B1 → 30

15 25 = Copy formula : X

formula → A2 + B2 → 40

Explanation: when the formula is copied down

A1 change to A2

B1 change to B2

② Absolute Cell Reference

absolute cell reference is a cell reference that does not change when we copy or drag the formula.

\$ ⇒ dollar

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Date _____
Page _____

EX:- multiply (5 is constant)

benefit of column D is fixed

A D B

approach 1: 5 is constant ← D is constant

1
2
3
5

so ⇒ \$D\$1

or A : 10

or 20

or 30

= A1 * \$D\$1 formula: 21 (or) 07 (or 21/24)

Ans:- A D B

1 5 5

2 5 10

3 5 15

4 5 20

5 5 25

Add : (5)

EX: A D B

1
2
3

25 = 5 + 20

20 = ⇒ column D is fixed & row is fixed

(10 + 10) with work : 20

= A1 + \$D\$1

Ans A D B

1 5 6

2 5 7

3 5 8

③ Mixed cell Reference :-

Either the row or the column is fixed.

EX: $\$A1 \rightarrow$ column A is fixed, row change.

EX:

A	B
10	20
20	40

① EX: Column fixed ($\$A1$) $\times 1A =$

A	B	C	D	E	F
10		2		1	
20		01		2	
30		71		0	
		05		2	

Formula = $\$A1 + 5$

A column fixed and row change hoti

Answer:-

$$A1 + 5 = 15$$

$$A2 + 5 = 25$$

$$A3 + 5 = 35$$

② EX: Row fix ($A\$1$)

A	B	C	D	E	F
10					
20					
30					

Formula = $A\$1 \times 2$

Row 1 fixed and column change hoti.

A

B

10

 $A * 2$ $\rightarrow 20$

20

40

30

60

Operator: +, -, *, /, %

Arithmetic operator: +, -, *, /, %

Operator Formula Ex (A=10, B=5) Result

+

Addition

 $= A + B$

15

-

Subtraction

 $= A - B$

5

*

Multiplication

 $= A * B$

50

/

Division

 $= A / B$

2

^

Power

 $= A ^ B$ $100 = (10^2)$

Comparison operators: =, <, >, <=, >=, <=, >=

=

Equal to

 $= A = B$ (9/10) False

>

Greater than

 $= A > B$ True

<

Less than

 $= A < B$ (9/10) False

>=

Greater than or equal to

 $= A >= B$ True

<=

Less than or equal to

 $= A <= B$ (9/10) False

<>

Not equal to

 $= A <> B$ True

Case Statement in Excel :-

IF Function :-

formula :- $\text{if}(\text{condition}, \text{value-if-true}, \text{value-if-false})$

Ex: Student Result :-

A (Name) B (marks) C (Result)

Rahul - 80 Pass

Priya - 35 fail

Ram - 40 Pass

Ans: formula use Result (Pass/fail)

C2 :- $\text{if}(B2 \geq 40, \text{"Pass"}, \text{"fail"})$

= $\text{if}(B2 \geq 40, \text{"Pass"}, \text{"fail"})$

Ans: $\Rightarrow$ O/P $\Rightarrow$ Rahul = 80 Pass

Ans: $\Rightarrow$ Priya = 35 fail

Ans: $\Rightarrow$ Ram = 40 Pass

Ans: $\Rightarrow$ Priya = 35 fail

Ans: $\Rightarrow$ Ram = 40 Pass